Microsoft Elevate

CAPSTONE PROJECT

Energy Trade Analysis - India

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Problem Statement:

India's increasing energy demand makes the country highly dependent on imported energy resources such as crude oil, coal, and natural gas. In recent years, global events such as geopolitical conflicts and disruptions in international trade have caused fluctuations in energy prices and supply. Understanding long-term trends in energy imports and exports is important to analyze India’s dependency on foreign energy and its vulnerability to global economic and political changes.

Proposed Solution:

The proposed system is an interactive Power BI dashboard designed to analyze India's energy trade data from 1970 to 2023. The system processes historical data related to major energy resources such as coal, crude oil, petroleum products, natural gas, electricity, and lignite.

The dashboard transforms complex trade datasets into meaningful visualizations, allowing users to easily observe long-term import trends, compare imports and exports across different energy sources, and identify the major contributors to India's energy imports.

The system includes KPI cards, trend analysis charts, comparison graphs, and distribution visuals, along with a year filter that enables users to explore data for specific time periods.

By presenting energy trade data in an interactive and visual format, the system helps users understand India's dependency on foreign energy resources and supports better analysis of energy trade patterns.

System Approach:

Tools and Technologies Used

- Power BI Desktop – Data visualization and dashboard creation
- NDAP (NITI Aayog Data Platform) – Data source for energy trade statistics
- Power Query – Data cleaning and transformation

Dataset Information

- Country: India
- Time period: 1970 – 2023
- Energy types analyzed:
  - Coal
  - Crude Oil
  - Petroleum Products
  - Natural Gas
  - Electricity
  - Lignite

Algorithm & Deployment:

Step 1 – Data Collection

Energy trade data is collected from the NDAP (NITI Aayog Data Platform).

Step 2 – Data Cleaning

The dataset is cleaned using Power Query to remove unwanted columns and extract year information.

Step 3 – Data Transformation

The data is structured into fields such as Year, Energy Type, Imports, Exports, and Net Imports.

Step 4 – Data Visualization

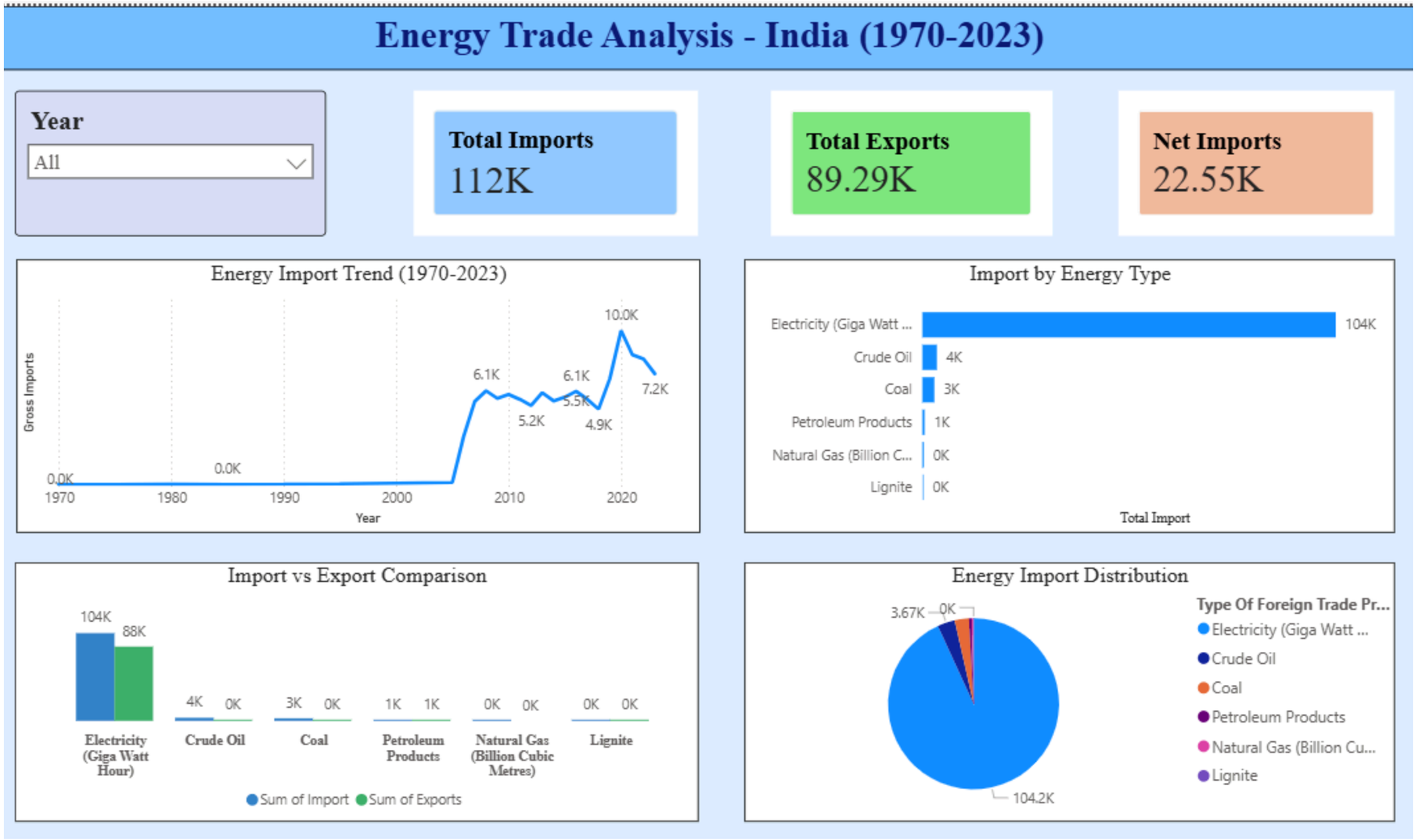
A Power BI dashboard is created with the following components:

- KPI Cards – Total Imports, Total Exports, Net Imports
- Line Chart – Energy Import Trend over time
- Bar Chart – Imports by Energy Type
- Column Chart – Import vs Export Comparison
- Pie Chart – Energy Import Distribution
- Slicer – Year filter for interactive analysis

Step 5 – Dashboard Deployment

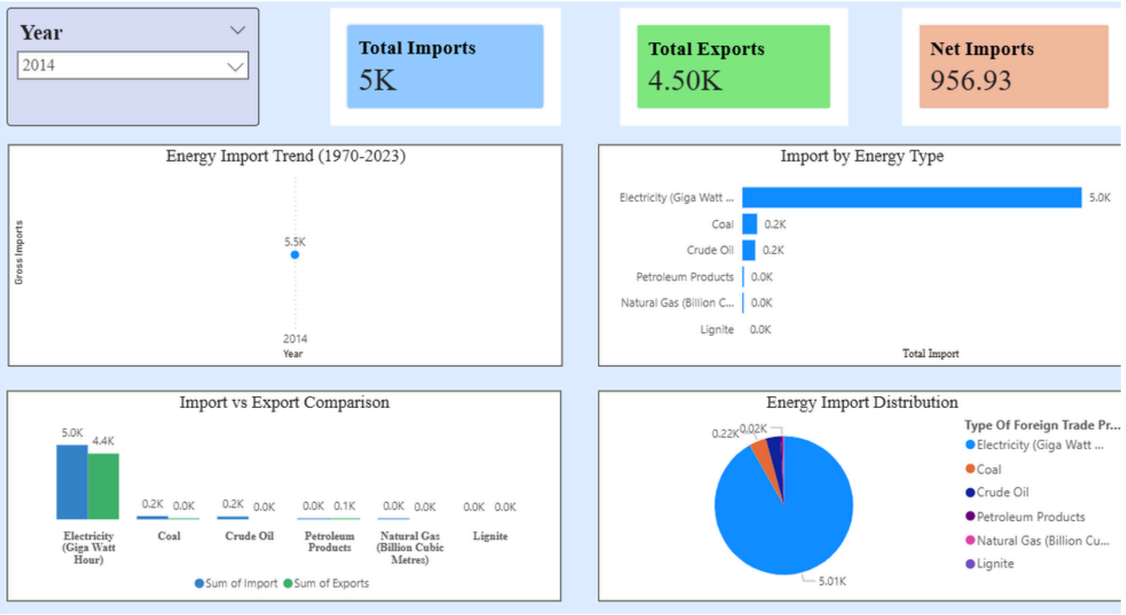
The final dashboard provides an interactive platform for analyzing India's energy trade patterns.

Result:



The Power BI dashboard successfully visualizes India's energy trade data from 1970 to 2023 using interactive charts and KPI indicators and silcer . It allows users to analyze trends in energy imports and exports across different energy sources such as coal, crude oil, natural gas, and electricity. The dashboard clearly shows a steady increase in energy imports over the years. These visual insights help users understand patterns in India's energy trade and dependency on external energy resources.

Example: Energy Trade Data of 2014



Energy Trade Data of 2022

The dashboard provides insights into:

- Total imports, exports, and net imports of energy resources
- Long-term trends in energy imports
- Comparison of imports and exports across different energy types
- Distribution of energy imports by product category

These visualizations help users quickly understand energy trade patterns and identify the major contributors to India's energy imports.

Conclusion:

The Energy Trade Analysis dashboard provides a clear visualization of India's energy import and export trends over the past five decades. The analysis shows that energy imports have increased significantly due to growing industrialization and energy demand. The results also highlight India's dependence on foreign energy sources, which can be affected by global geopolitical events and international market fluctuations.

The project demonstrates how Power BI can be used to transform raw energy trade data into meaningful visual insights. The analysis shows that India's energy imports have increased significantly due to growing industrial and population demands. The dashboard helps in understanding long-term energy trade trends and supports better analysis for future energy planning and policy decisions.

Future scope:

In the future, this project can be extended by integrating real-time energy trade and global fuel price data to provide more up-to-date analysis. Advanced analytics techniques such as machine learning models can also be used to forecast future energy import trends and demand patterns.

The dashboard can further include renewable energy data such as solar, wind, and hydro to analyze India's transition toward sustainable energy sources. Additional features like state-wise energy consumption analysis and predictive insights can help policymakers and researchers make more informed decisions regarding energy planning and national energy security.

References:

- NDAP – NITI Aayog Data Platform.
- Microsoft Power BI Documentation.
- Government of India – Energy Statistics Reports
- GitHub: https://github.com/riyaau6816-ship-it/India-Energy-Trade-Analysis-PowerBI-ru

Thank You